10/07/2020

CODE-A

Aakash
Medical | IIT-JEE | Foundations
(Divisions of Aakash Educational Services Limited)
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Aakash Rank Booster Test Series for NEET-2020

MM : 720 **Test - I8** **Time : 3.00 Hrs.**

Topics covered : Complete Syllabus of Class XI and XII.

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. A car travels half of its destination with speed of 40 km/h and another half with speed *v*. If the average speed of the car is 48 km/h then the value of *v* is
   (1) 55 km/h (2) 60 km/h
   (3) 52 km/h (4) 50 km/h

2. In a case of a projectile projected from ground, the velocity is perpendicular to acceleration at
   (1) Highest point of projection
   (2) Point of project
   (3) Point of landing
   (4) Point equal half of maximum height

3. A satellite is orbiting very close to surface of the earth. Its time period is approximately
   (1) 42.3 minutes (2) 4.4 hours
   (3) 2.4 hours (4) 84.6 minutes

4. Which of the following is a unit vector?
   (1) $\frac{\hat{i}}{\sqrt{2}}$ (2) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$
   (3) $\frac{\hat{j}}{\sqrt{2}}$ (4) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{2}}$

5. If $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$, then the angle between vector $\vec{A}$ & $\vec{B}$ is
   (1) 60° (2) 30°
   (3) 90° (4) 45°

6. In uniform circular motion which of the following quantity is zero?
   (1) Change in acceleration
   (2) Magnitude of change in velocity
   (3) Change in magnitude of velocity
   (4) Magnitude of change in linear momentum

7. Which of the following statement is correct for friction?
   (1) Static friction is self adjusting friction
   (2) Kinetic friction is self adjusting friction
   (3) Limiting friction is self adjusting friction
   (4) Both (1) & (2)

8. The change in surface energy when a drop of radius *R* splits into 8 droplets, each of radius *r* is (surface tension is *T*)
   (1) $2\pi R^2T$ (2) $4\pi R^2T$
   (3) $8\pi R^2T$ (4) πR^2T

9. Which of the following relation is correct? (Where symbol have their usual meanings)

(1) $\vec{F} = \vec{\tau} \times \vec{r}$ (2) $\vec{\tau} = \vec{r} \times \vec{F}$

(3) $\vec{F} = \vec{r} \times \vec{\tau}$ (4) $\vec{\tau} = \vec{F} \times \vec{r}$

10. A body cooling from 70°C to 50°C in 1 minute. The time taken to cool same body from 50°C to 40°C is (Take surrounding temperature is 30°C)

(1) 1 minute

(2) Less than 1 minute

(3) Cannot be predicted

(4) Greater than 1 minute

11. Which of the following gates corresponds to the given truth table?

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0

(1) NAND (2) NOR

(3) XOR (4) OR

12. Which of the following regions of output characteristic of transistor is used when it is used as amplifier?

(1) Saturation region (2) Cut off region

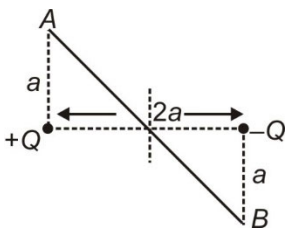
(3) Active region (4) All of these

13. In hydrogen atom, for which transition minimum energy is emitted?

(1) $n_1 = 2, n_2 = 3$ (2) $n_1 = 2, n_2 = 4$

(3) $n_1 = 3, n_2 = 4$ (4) $n_1 = 4, n_2 = 5$

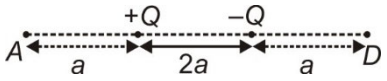
14. The potential difference between two points *A* and *B* is



(1) Zero (2) $\frac{Q}{4\pi\varepsilon_0 a}\left(1-\frac{1}{\sqrt{5}}\right)$

(3) $\frac{Q}{2\pi\varepsilon_0 a}\left[1-\frac{1}{\sqrt{5}}\right]$ (4) $\frac{Q}{4\pi\varepsilon_0 a}\left[2-\frac{1}{\sqrt{5}}\right]$

15. What is the work done by the electric field to take any charge *q* from point *A* to *D*?



(1) Zero (2) $\frac{2}{3}\frac{Qq}{4\pi\varepsilon_0 a}$

(3) $\frac{Qq}{34\pi\varepsilon_0 a}$ (4) $\frac{4}{3}\frac{Qq}{4\pi\varepsilon_0 a}$

16. In an ideal thermodynamic system, gas is allowed to expand from volume V_1 to V_2. The work done by the system is maximum if it is

(1) Isothermal expansion

(2) Adiabatic expansion

(3) Isobaric expansion

(4) Work done will be same for all processes

17. Equation of a plane progressive harmonic wave is given as $y = 0.02 \sin\left(20t - \frac{1}{25}x + \frac{\pi}{6}\right)$. The speed of wave is (All quantities are in SI units)

(1) $\frac{5}{4}$ m/s (2) 250 m/s

(3) 500 m/s (4) 750 m/s

18. If two identical wires have resistivity ρ_1 and ρ_2. They are connected in series, then the equivalent resistivity is equal to

(1) $\rho_1 \rho_2$ (2) $\rho_1 + \rho_2$

(3) $\frac{\rho_1 + \rho_2}{2}$ (4) $\frac{\rho_1 - \rho_2}{2}$

19. A wire of resistance 20 Ω is bent to form a square. What will be the equivalent resistance between diagonally opposite points?

(1) 10 Ω (2) 5 Ω

(3) 15 Ω (4) 2.5 Ω

20. A ball is moving with velocity 5 m/s towards a heavy wall moving towards the ball with speed 2 m/s. Assuming collision to be elastic, what will be the speed of ball immediately after the collision?

(1) 9 m/s (2) 3 m/s

(3) 2.5 m/s (4) 5 m/s

21. A force $\vec{F} = \frac{-k}{x^2}\hat{i}\,(x \neq 0)$ acts on a particle moving in *x*-direction. The work done by this force in displacing a particle from $x = +a$ to $x = +2a$ (k = Positive constant) is equal to

(1) $\frac{-k}{2a}$ (2) $\frac{k}{2a}$

(3) $\frac{k}{a}$ (4) $\frac{-k}{a}$

22. In a velocity selector, electric field vector and magnetic field vector must be oriented such that angle between them is

(1) 0° (2) 90°

(3) 180° (4) 360°

23. What will be the moment of inertia of a uniform square plate of mass *M* and side *a* about an axis perpendicular to plane passing through its centre?

(1) $\frac{Ma^2}{6}$ (2) $\frac{Ma^2}{12}$

(3) $\frac{Ma^2}{8}$ (4) $\frac{2Ma^2}{5}$

24. The principle of working of transformer is

(1) Self induction (2) Mutual induction

(3) Bernoulli's principle (4) All of these

25. What is the pressure head in Bernoulli's theorem?

(p = pressure, ρ = density, g = acceleration due to gravity)

(1) $\frac{p}{\rho}$ (2) $\frac{gp}{\rho}$

(3) $\frac{p}{\rho g}$ (4) p

26. A biconvex lens has radii of curvature 10 cm each. If the refractive index of the material of lens is 1.5, then the focal length of lens is

(1) 20 cm (2) 10 cm

(3) – 20 cm (4) – 10 cm

27. Keeping all parameter fixed if distance between two slits in Young's double slit experiment is decreased, then fringe width will

(1) Increase

(2) Decrease

(3) Remain same

(4) May increase or decrease depends on intensity of source

28. Which of the following relations is a correct Einstein's photoelectric equation? [All symbols have their usual meaning]

(1) $K_{max} = h\nu - \phi_0$ (2) $eV_0 = \frac{hc}{\lambda} - \phi_0$

(3) $V_0 = \frac{hc}{e}\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$ (4) All of these

29. In Bohr's atomic model, the time period of an electron in n^{th} orbit depends on

(1) $T_n \propto \frac{n^3}{Z^2}$ (2) $T_n \propto \frac{n^3}{Z^3}$

(3) $T_n \propto \frac{n^2}{Z^2}$ (4) $T_n \propto \frac{n}{Z}$

30. A magnet of length '*L*' and magnetic moment *M* is given. If it is bent in the shape as shown in figure, then the new magnetic moment will be



(1) $\frac{M}{2}$ (2) $\frac{2M}{3}$

(3) $\frac{2M}{3\pi}$ (4) $\frac{2M}{\pi}$

31. A PN junction diode is made up of a material which has energy gap of 3.0 eV. The maximum wavelength or radiation that can be absorbed by the material is nearly equal to

(1) 6216 Å (2) 5426 Å

(3) 4140 Å (4) 3146 Å

32. An electron (rest mass = m_0) is moving with speed v so that its mass becomes $\frac{5}{3}m_0$, then the value of v is

(1) $0.5c$ (2) $\frac{c}{5}$

(3) $\frac{2}{5}c$ (4) $\frac{4}{5}c$

33. When $_{92}U^{237}$ nucleus is converted into $_{83}Bi^{209}$, x and y number of α and β^- particles are emitted respectively then the value of x and y are respectively

(1) 5, 7 (2) 7, 6

(3) 7, 5 (4) 6, 7

34. The longest wavelength emitted in hydrogen atom belonging to Balmer series, is

(1) $\frac{16}{3R}$ (2) $\frac{36}{7R}$

(3) $\frac{16}{7R}$ (4) $\frac{36}{5R}$

35. If a radiation of wavelength 200 nm is incident on a metal surface having work function 4 eV, then the value of stopping potential required is

(1) 1.08 V (2) 2.21 V

(3) 3.52 V (4) 6.21 V

36. Ratio of intensities of two waves from slits S_1 and S_2 is 4 : 1 respectively. Then the ratio of maximum to that of minimum intensities in YDSE is
(1) 3 : 1 (2) 9 : 1
(3) 9 : 5 (4) 3 : 2

37. If angle of minimum deviation by an equilateral prism is equal to its prism angle, then the refractive index of material of prism can be
(1) 1.12 (2) 1.73
(3) 1.40 (4) 1.08

38. An air bubble in a glass slab (μ = 1.5) appears 3 cm deep when viewed from one face and 4 cm when viewed from opposite face, then real thickness of glass slab is
(1) 8.5 cm (2) 9 cm
(3) 9.5 cm (4) 10.5 cm

39. A transformer is used to illuminate a bulb rated as 100 W and 110 V, from a 220 V mains. If main current is 0.6 A, then the efficiency of transformer is nearly
(1) 100% (2) 76%
(3) 88% (4) 94%

40. 40 W, 220 V and 110 W, 220 V bulbs are connected in series with 440 volt supply, then
(1) 110 W bulb will fuse
(2) 40 W bulb will fuse
(3) Both 110 W and 40 W will fuse
(4) No bulb will fuse

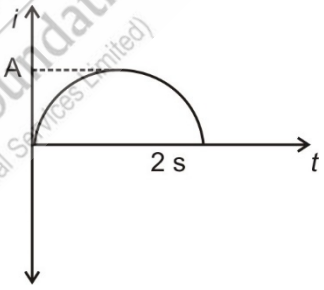
41. When plane of dip circle is in magnetic meridian it shows dip angle 30°, if plane of dip circle is turned by 30°, the new dip angle is
(1) $\tan^{-1}\left(\frac{3}{2}\right)$ (2) $\tan^{-1}\left(\frac{2}{3}\right)$
(3) $\tan^{-1}(2)$ (4) $\tan^{-1}\left(\frac{5}{3}\right)$

42. The instantaneous angular position of a point on rotating wheel is given as $\theta = 4t^3 - 12t^2$. The torque on the wheel is zero at
(1) 1 s (2) 1.5 s
(3) 2 s (4) 2.5 s

43. When a bob is suspended by light string and it is imparted a horizontal velocity to complete the full circular trajectory in vertical plane. What is the difference in tension between lowest and highest point of path?
(1) mg
(2) $6mg$
(3) $5mg$
(4) $3mg$

44. A monatomic gas at 300 K is compressed adiabatically to $\frac{1}{8}$ th of its initial volume. The final temperature of the gas is
(1) 1200°C (2) 927°C
(3) 527°C (4) 527 K

45. If the current flowing in a conductor varies with time as shown (in a semi-circular path), then find the net charge flown in 2 seconds



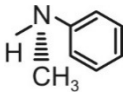
(1) 1.57 C
(2) 2.57 C
(3) 3.14 C
(4) 0.57 C

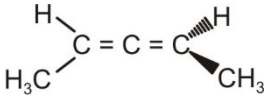
CHEMISTRY

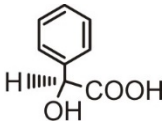
46. Which metal carbonyl act as better oxidizing agent?
(1) $Ni(CO)_4$ (2) $Cr(CO)_6$
(3) $Fe(CO)_5$ (4) $V(CO)_6$

47. Among the following which is most likely unhybridized molecule?
(1) PCl_3 (2) PCl_5
(3) PH_3 (4) NH_3

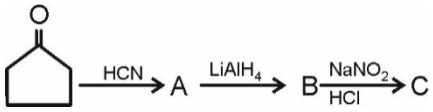
48. Among the following, the optically inactive compound is

(1) 

(2) 

(3) 

(4) Both (2) and (3)

49. 

Number of lone pair present in compound (C), obtained in above given reactions mechanism, is/are

(1) 2

(2) 3

(3) 4

(4) 1

50. Which is not correct match in the following?

(1) 0.2 gm molecules $C_6H_{12}O_6$ – 2.4 gm atoms H

(2) 0.1 mol $H_2(g)$ (at one atm pressure and 273°C temperature) – 4.08 L gas

(3) 6.022×10^{23} atoms hydrogen – 0.1mole $C_4H_{10}(g)$

(4) 50 ml Hg (density = 13.6 gm/ml) (atomic mass Hg : 200 u) – 3.4 gm atoms Hg

51. Choose the **incorrect** order as the property indicated

(1) $NH_3 > SbH_3 > AsH_3 > PH_3$ (Boiling point)

(2) $NH_3 > SbH_3 > AsH_3 > PH_3$ (Melting point)

(3) $NH_3 > PH_3 > AsH_3 > SbH_3$ (Bond angle)

(4) N–H < P–H < As–H < Sb–H (Bond length)

52. Which of the following species is diamagnetic in nature?

(1) KO_2 (2) Na_2O_2

(3) $O_2[PF_6]$ (4) ClO_2

53. Which of the following molecule has non-zero dipole moment?

(1) PCl_2F_3

(2) XeF_2

(3) I_3^-

(4) XeF_4

54. Incorrect order of thermal stability is

(1) $BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3$

(2) $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2$

(3) MgO < CaO < SrO < BaO

(4) $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4$

55. Which of the following will not form disproportionated product?

(1) $Cl_2 + \underset{(dil)}{NaOH} \xrightarrow{Cold}$

(2) $Br_2 + \underset{(conc)}{NaOH} \xrightarrow{Hot}$

(3) $XeF_4 + H_2O \longrightarrow$

(4) $XeF_2 + H_2O \longrightarrow$

56. Which of the following metal nitrate may decomposes into corresponding nitrites on heating?

(1) $NaNO_3$ (2) $LiNO_3$

(3) $Mg(NO_3)_2$ (4) $Ba(NO_3)_2$

57. Dry ice is

(1) $CO_2(s)$ (2) $H_2O(s)$

(3) $D_2O(s)$ (4) $H_2O_2(s)$

58. The total number of voids in face centered cubic unit cell is

(1) 4 (2) 8

(3) 10 (4) 12

59. Consider the following statements.

(i) On adding methyl alcohol to water, boiling point of water increases

(ii) When NaCl is added to water, depression in freezing point is observed

(iii) Diffusion of solution occurs from a region of its high concentration to a region of its low concentration

Choose the **correct** options

(1) (i) and (ii)

(2) (i) and (iii)

(3) (ii) and (iii)

(4) (i), (ii) and (iii)

60. What is the value of ΔG° for the following cell reaction?

$Zn(s) + Ag_2O(s) + H_2O(l) \rightarrow Zn^{2(+)}(aq) + 2Ag(s) + 2OH^{(-)}(aq)$

$\left[\text{Given: } E^\circ_{Zn/Zn^{2(+)}} = +0.76\,V, E^\circ_{Ag_2O/Ag} = +0.80\,V\right]$

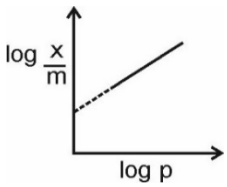
(1) -1.9×10^7 kJ

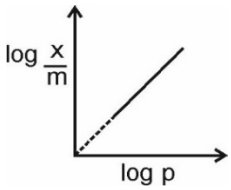
(2) – 301 kJ

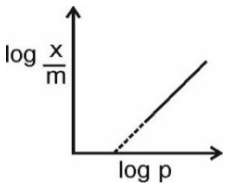
(3) -1.8×10^3 kJ

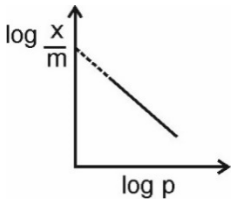
(4) – 200 kJ

61. Which of the following is the **correct** plot for Freundlich adsorption isotherm?

(1) 

(2) 

(3) 

(4) 

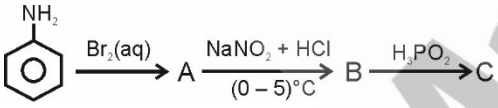
62. Iron is galvanised by
(1) Cu (2) Zn
(3) Au (4) Ag

63. Consider the following complex ions

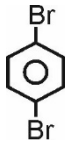
$[CrCl_6]^{3-}$, $[Cr(NH_3)_6]^{3+}$, $[Cr(CN)_6]^{3-}$
I II III

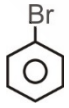
The **correct** order of crystal field splitting energy is
(1) I > II > III (2) II > III > I
(3) III > II > I (4) I > III > II

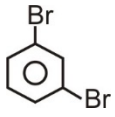
64. The number of geometrical isomers possible for the complex of the type $[Ma_2b_2c_2]$ is
(1) 4 (2) 3
(3) 5 (4) 8

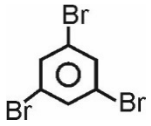
65. 

The compound (C) is

(1) 

(2) 

(3) 

(4) 

66. The reductive ozonolysis of natural rubber will finally form
(1) Acetone and Ethanal
(2) 4-Oxopentanal
(3) Acetone and Propanal
(4) All of these

67. Which of the following is a biodegradable polymer?
(1) Nylon–6, 6 (2) Nylon–2–nylon–6
(3) Nylon–6 (4) All of these

68. How much electricity in terms of faraday is required to produce 3 g of Al from molten Al_2O_3?
(1) 1 F (2) 0.5 F
(3) 0.33 F (4) 0.22 F

69. For a first order reaction, the time required for 99.9% completion is 100 min. What will be the rate constant for the reaction?
(1) $1.25 \times 10^{-4}\ s^{-1}$ (2) $1.15 \times 10^{-3}\ s^{-1}$
(3) $2.2 \times 10^{-2}\ s^{-1}$ (4) $4.5 \times 10^{-2}\ s^{-1}$

70. Consider the following statements
a. Molecularity of a reaction can be zero.
b. Order of a reaction can be fractional.
c. Molecularity is not defined for a complex reaction.

Choose the incorrect statement(s)
(1) a only (2) a and c only
(3) b and c only (4) a, b and c

71. Which of the following is not an example of heterogeneous catalysis?
(1) $2SO_2(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$
(2) $N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$
(3) $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$
(4) $4NH_3(g) + 5O_2(g) \xrightarrow{Pt(s)} 4NO(g) + 6H_2O(l)$

72. Consider the following statement.
a. H_2S is less acidic than H_2Te
b. PCl_3 undergoes hydrolysis to give fumes of HCl
c. PH_3 is basic as it forms phosphonium iodide with HI
d. PH_3 has higher boiling point that NH_3

Choose the correct statements
(1) a, b and c only (2) b, c and d only
(3) a, c and d only (4) a, b, c and d

73. Choose the **correct** statement among the following
(1) OF_2 is known as oxygen difluoride because F is less electronegative than oxygen
(2) SO_2 is both oxidising as well reducing but SO_3 is only oxidising
(3) SF_4 can act as Lewis base and also as Lewis acid
(4) Ozone is a strong oxidising agent but less powerful than O_2

74. Which of the following is the correct match?

(1) Extent of hydration : $Cl^- > F^- > Br^- > I^-$

(2) Boiling point : $F_2 > Cl_2 > Br_2 > I_2$

(3) Bond energy : $Cl_2 > F_2 > Br_2 > I_2$

(4) Bond length : $I_2 > Br_2 > Cl_2 > F_2$

75. Identify the ion which should be colourless in aqueous medium

(1) V^{3+} (2) Mn^{2+}

(3) Co^{2+} (4) Sc^{3+}

76. The complex in which Ni is sp^3 hybridized with +2 oxidation state is

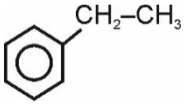
(1) $[Ni(CO)_4]$ (2) $[NiCl_4]^{2-}$

(3) $[Ni(CN)_4]^{2-}$ (4) All of these

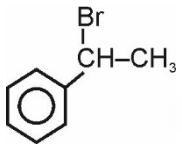
77. Which of the following complexes is optically active?

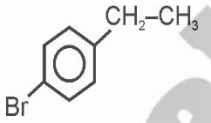
(1) $[MA_2B_4]$ (2) $[MA_3B_3]$

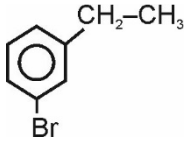
(3) $[M(AA)_3]$ (4) All of these

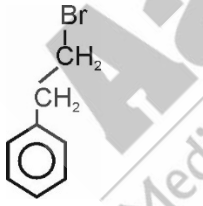
78.  $\xrightarrow[h\nu]{Br_2}$ X(Major).

The major product X is

(1) 

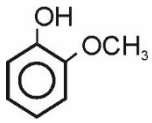
(2) 

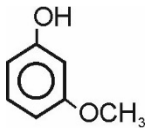
(3) 

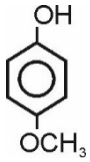
(4) 

79. Arrange the following in the correct order of acidic strength

(a) 

(b) 

(c) 

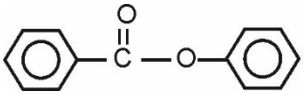
(d) 

(1) (c) > (b) > (a) > (d)

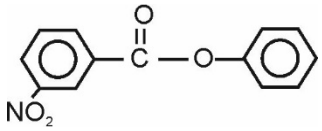
(2) (a) > (b) > (c) > (d)

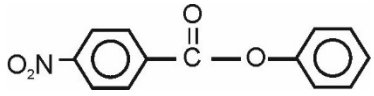
(3) (a) > (c) > (b) > (d)

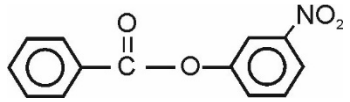
(4) (a) > (d) > (b) > (c)

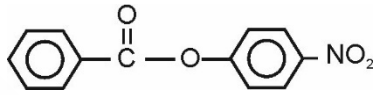
80. 

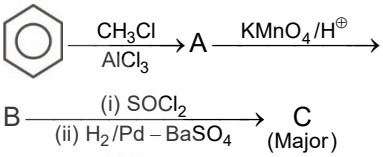
The compound 'Z' is

(1) 

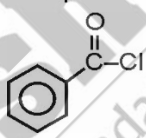
(2) 

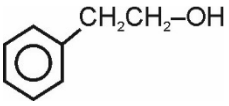
(3) 

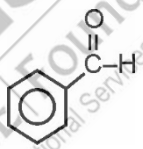
(4) 

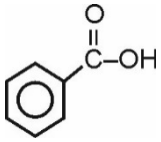
81. 

The compound 'C' is

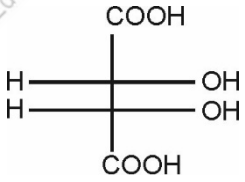
(1) 

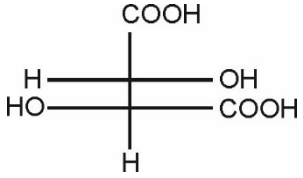
(2) 

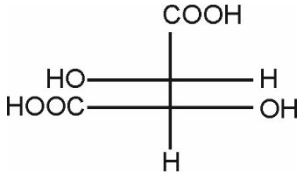
(3) 

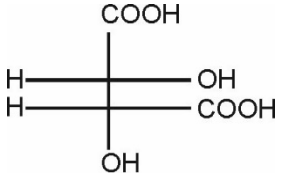
(4) 

82. The compound which is not meso-tartaric acid

(1) 

(2) 

(3) 

(4) 

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83. Choose the incorrect match
(1) Vitamin K : Increased blood clotting time
(2) Vitamin D : Cheilosis
(3) Vitamin C : Bleeding gums
(4) Vitamin B_{12} : Pernicious anaemia

84. The empirical formula of a compound is CH_2O and its molecular mass is 90 g. The molecular formula of the compound is
(1) $C_4H_8O_4$ (2) $C_3H_6O_3$
(3) $C_6H_{12}O_6$ (4) $C_2H_4O_2$

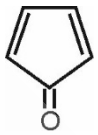
85. The maximum number of electrons possible in an atom having (n + l) value equal to 3, is
(1) 2 (2) 6
(3) 8 (4) 10

86. K_a for CH_3COOH is 10^{-5} and K_b for NH_4OH is 10^{-5}. The pH value of ammonium acetate will be
(1) 10 (2) 5
(3) 7 (4) 7.5

87. Oxidation state of Cr in CrO_5 is
(1) +10 (2) +8
(3) +6 (4) +5

88. The 'Calgon' is
(1) $Na_6P_6O_{18}$ (2) $Na_2S_2O_6$
(3) $NaAlSi_2O_8S$ (4) $CaSO_4.MgSO_4$

89. Number of B–O–B linkages present in borax is
(1) 2 (2) 5
(3) 1 (4) Zero

90. The compound which is aromatic is
(1) 
(2) 
(3) 
(4) 

BOTANY

91. A are consumer decomposer protists. Under favorable conditions, they form B which may grow and spread over several feet. Choose the correct option for A and B.

	A	B
(1)	Slime moulds	Fruiting bodies
(2)	Euglenoids	Plasmodium
(3)	Slime moulds	Plasmodium
(4)	Euglenoids	Fruiting bodies

92. How many of the following are **incorrectly** matched?
(a) *Porphyra* – chlorophyll a and d
(b) *Dictyota* – Fucoxanthin
(c) *Ulothrix* – Mannitol
(d) *Sargassum* – Laminarin
(1) 1 (2) 3
(3) 4 (4) 2

93. Identify the organism that shows the following characteristics.
(a) Zygotic meiosis
(b) Causes red tide of the sea
(c) Mesokaryon organisation
(d) Whirling whips
(1) *Noctiluca* (2) *Gymnodinium*
(3) *Melosira* (4) *Navicula*

94. Select the **mismatched** pair.

Fruits	Edible parts
(1) Cashew nut	Cotyledons
(2) Mango	Fleshy mesocarp
(3) Almond	Mesocarp
(4) Apple	Fleshy thalamus

95. Read the following statements and identify the **incorrect** ones.
(a) Intrafascicular and interfascicular cambium are the lateral meristems that participate in the secondary growth of dicot stem.
(b) The heartwood comprises dead elements with highly lignified cell walls.
(c) Phelloderm has thin walled parenchymatous cells.
(d) The oldest layer of phloem lies just outside the pericycle.
(e) Phloem parenchyma is absent in most of the monocots.
(1) All except (d) (2) (a), (b) and (c)
(3) (d) only (4) (c) and (e)

96. Identify the stages of cell cycle in which the following events occur.

(A) Duplication of most of the organelles

(B) Duplication of centrioles

(C) Synthesis of tubulin proteins

	(A)	(B)	(C)
(1)	G_2 phase	G_1 phase	M phase
(2)	S phase	M phase	G_0 phase
(3)	M phase	Interphase	G_1 phase
(4)	G_1 phase	S phase	G_2 phase

97. Match the following columns

	Column-I	Column-II
a.	Mg^{2+}	(i) Alcohol dehydrogenase
b.	Mo	(ii) RuBisCO
c.	Zn^{2+}	(iii) Catalase
d.	Iron	(iv) Nitrogenase

(1) a(iv), b(ii), c(i), d(iii)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(ii), b(iv), c(iii), d(i)

(4) a(iii), b(i), c(ii), d(iv)

98. In C_4 plants, number of carbon atoms in

(a) Primary CO_2 acceptor

(b) Primary CO_2 fixation product

	(a)	(b)
(1)	5	3
(2)	3	4
(3)	3	5
(4)	4	3

99. All of the enzymes of Krebs' cycle are found in mitochondrial matrix, **except**

(1) α-ketoglutaric dehydrogenase complex

(2) Malic dehydrogenase

(3) Fumarase

(4) Succinate dehydrogenase

100. **Statement A**: Shuttles do not occur in prokaryotes.

Statement B: $NADH_2$ molecules produced during glycolysis can directly enter the mitochondria.

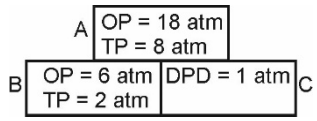
(1) Only A is correct

(2) Only B is correct

(3) Both A and B are correct

(4) Both A and B are incorrect

101. Find the direction of water flow in the given system



(1) A $\longrightarrow$ C (2) B $\longrightarrow$ C

(3) C $\longrightarrow$ A (4) A $\rightleftharpoons$ B

102. All the given plants are dioecious, **except**

(1) Papaya (2) Date palm

(3) *Marchantia* (4) Coconut

103. Choose the **incorrect** statement w.r.t. pollen grain.

(1) Exine of the pollen grain has prominent apertures called germ pores where sporopollenin is absent

(2) Inner wall of the pollen grain is called intine

(3) Pollen grains are generally spherical in shape

(4) In 60 percent of angiosperms, pollen is shed at three-celled stage

104. The pure line round seeded pea plant was crossed with wrinkled seeded pea plant. The F_1 generation is _____ and it can be explained by _____, (respectively)

(1) Wrinkled ; Law of segregation

(2) Round ; Law of dominance

(3) Round ; Codominance

(4) Wrinkled ; Law of dominance

105. "Pusa Swarnim" is a/an

(1) Aphid resistant variety of *Brassica*

(2) White rust resistant variety of *Brassica*

(3) Shoot and fruit borer resistant variety of flat bean

(4) Hill bunt resistant variety of wheat

106. In STP

(1) Primary treatment is a biological process

(2) Flocs are anaerobic and autotrophic microbes

(3) BOD measures the organic matter present in the water

(4) Activated sludge is the effluent of tertiary treatment

107. "Sexual deceit" is employed by

(1) *Ophrys* to get pollinated

(2) Orchids on mango for shelter

(3) Cuckoo on crow

(4) Abingdon tortoise in Galapagos island

108. Humus is

A. Dark coloured amorphous substance

B. Highly resistant to microbial action

(1) Both A & B are correct

(2) Both A & B are incorrect

(3) A is correct but B is incorrect

(4) A is incorrect but B is correct

109. Most important cause for biodiversity loss is

(1) Over exploitation

(2) Alien species invasion

(3) Co-extinctions

(4) Habitat loss and fragmentation

110. The international treaty to control ozone depletion is

(1) Kyoto Protocol (2) Montreal Protocol

(3) Bali convention (4) Earth Summit

111. Read the following statements and choose the **correct** pair.

(1) Collection of preserved plant and animal specimens – Herbarium

(2) *Solanum*, *Panthera*, *Triticum*, *Felis* – Genera

(3) Reproduction is synonymous to growth – *Albugo*

(4) Study of diversity of living organisms along with their evolutionary relationships – Taxonomy

112. Fill in the blanks and choose the **correct** option.

A. Mycelium is coenocytic and aseptate in class _____.

B. _____ undergoes rapid multiplication and causes red tide

C. Sleeping sickness is caused by _____ protozoan.

D. The protein coat of virus consist of small subunits called _____.

	(i)	(ii)	(iii)	(iv)
(1)	Ascomycetes	Cyanobacterium	Flagellated	Peplomeres
(2)	Phycomycetes	*Gonyaulax*	Flagellated	Capsomeres
(3)	Basidiomycetes	Chrysophyte	Amoeboid	Capsomeres
(4)	Deuteromycetes	Euglenoid	Cilliated	Peplomeres

113. W.M. Stanley

(1) Discovered TMV

(2) Coined the term virus

(3) Demonstrated that extract of infected plant can cause tobacco mosaic disease in healthy plants

(4) Crystallised virus and showed that crystals are largely made up of proteins

114. Read the following and choose the **incorrect** statement.

(1) Majority of red algae are marine with greater concentrations found in the warmer areas

(2) Thallus of *Marchantia* is dorsiventral and closely appressed to the substrate

(3) In *Sequoia*, reduced male gametophyte is called pollen grain and its endosperm represents future sporophytes

(4) *Wolfia* is the smallest flowering plant

115. Which of the following is **not** related to the floral formula of belladonna?

(1) $\overset{\frown}{C_{(5)}\ A_{(5)}}$ (2) $\underline{G}_{(2)}$

(3) $\oplus\ ♀^{♂}$ (4) $K_{(5)}$

116. (i) cambium forms complete, continuous but wavy ring during secondary growth in (ii) .

(1) (i) Vascular (ii) Dicot stem (2) (i) Cork (ii) Dicot leaf

(3) (i) Vascular (ii) Dicot root (4) (i) Cork (ii) Dicot root

117. Read the following statements w.r.t. prokaryotic cell.

I. Single envelope system present.

II. Mesosome increases surface area for enzyme action.

III. The small circular DNA outside the genomic DNA confers unique phenotype.

IV. Contains pseudo-vacuoles.

(1) Only I is correct

(2) Only I and II are correct

(3) Only I, II and III correct

(4) I, II, III and IV are correct

118. Crossing over

(1) Occurs in dyad stage of meiosis

(2) Causes recombination between genes of homologous chromosomes

(3) Occurs after chiasmata formation in leptotene

(4) Occurs in diplotene, the longest phase of meiosis

119. Loss of water in liquid phase from the margin of leaves in many herbaceous plants is/occurs

(1) Guttation

(2) Transpiration

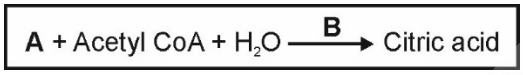
(3) At noon

(4) Due to cohesion-tension model of water movement

120. The essential element which maintains the ribosome structure is
(1) Responsible for carbohydrate translocation
(2) Needed during mitotic spindle formation
(3) Constituent of ring structure of chlorophyll
(4) Needed in the synthesis of auxin

121. Read the following statements.
Statement-I : Water splitting complex is located on the inner side of the thylakoid membrane.
Statement-II : During chemiosmosis breakdown of proton gradient releases energy.
(1) Both I and II are incorrect
(2) I is correct but II is incorrect
(3) I is incorrect but II is correct
(4) Both I and II are correct

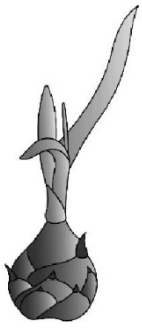
122. Complete the given reaction of the cycle present in mitochondrial matrix



(1) A–Succinic acid; B– Pyruvate dehydrogenase
(2) A–Malic acid; B–Transacetylase
(3) A–OAA; B–Citrate synthase
(4) A–α-KGA; B–Thiokinase

123. The phytohormone which was first isolated from human urine, shows
a. Bolting effect
b. Apical dominance
c. Control of xylem differentiation
d. Promote leaf senescence
e. Ripening of fruit
f. Initiation of rooting in stem cutting
(1) b, c and f (2) a, d and e
(3) b, e and f (4) a, b and d

124. The given vegetative propagule is

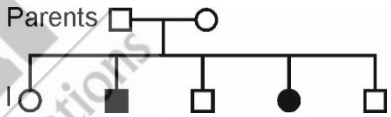


(1) Bulb of onion
(2) Rhizome of turmeric
(3) Tuber of potato
(4) Bulbil of *Agave*

125. Choose the **correct** statements
a. Perisperm is the persistant nucellus in the seeds of black pepper.
b. Single large cotyledon in grass is situated towards lateral side of the embryonal axis.
c. Seeds are the basis of agriculture.
d. *Citrus* has only one embryo in a seed.
(1) Only a
(2) Only a and b
(3) Only a, b and c
(4) a, b, c and d

126. A man suffering from haemophilia marries a normal woman without any history of haemophilia in her family. What percentage of their sons are affected?
(1) 100% (2) 75%
(3) 50% (4) 0%

127. Study the given pedigree chart and choose the **correct** statement.



(1) The trait under study is dominant
(2) Both parents are homozygous
(3) The trait can be X-linked recessive disease haemophilia
(4) The trait depicted is autosomal recessive like cystic fibrosis

128. In prokaryotes, semi-conservative DNA replication was first proved by using
(1) Radioactive ^{35}S and ^{32}P isotope
(2) Heavy ^{15}N isotope
(3) *Vicia faba*
(4) ^{3}H isotope in thymine

129. (i) The VNTR belongs to a class of satellite DNA referred to as __A__ .
(ii) Regulation of *lac* operon by repressor is referred to as __B__ regulation.
(iii) The secondary structure of tRNA looks like __C__ .

	A	B	C
(1)	Minisatellite	Positive	L-shaped
(2)	Micro-satellite	Negative	Inverted L- shaped
(3)	Minisatellite	Negative	Clover-leaf
(4)	Micro-satellite	Positive	Clover-leaf

130. In mung bean, resistance to yellow mosaic virus is induced by
(1) Mutation breeding (2) Hybridisation
(3) Selection (4) Polyploid breeding

131. The microbial biocontrol agent with species-specific, narrow spectrum insecticidal applications is
(1) *Trichoderma*
(2) *Glomus*
(3) Dragonflies
(4) *Nucleopolyhedrovirus*

132. Which of the following factors increases population density more significantly if new habitat is colonised?
(1) Natality (2) Mortality
(3) Immigration (4) Emigration

133. The natural reservoir for phosphorus is
(1) Ocean (2) Rock
(3) Atmosphere (4) Living organisms

134. "Hotspots of biodiversity", are/show
(1) High degree of endemism
(2) Slow rate of extinction
(3) Less species richness
(4) Only five in India

135. Natural aging of lake by biological enrichment of its water is
(1) Eutrophication
(2) Biomagnification
(3) Biofortification
(4) Bioprospecting

ZOOLOGY

136. Endometrial thickening in the follicular or proliferative phase is mainly brought about by
(1) Inhibin (2) Progesterone
(3) Estrogen (4) LH

137. Among the following which contraceptive method has the least failure rate?
(1) Coitus Interruptus (2) Barrier methods
(3) Oral contraceptives (4) Implants

138. Human papilloma virus is responsible for causing
(1) Syphilis (2) Genital herpes
(3) Chancroid (4) Genital warts

139. Stress leads to increase in all processes in the body, **except**
(1) Gluconeogenesis (2) Proteolysis
(3) Lipolysis (4) Glycogenesis

140. HIV infection can be confirmed by which test?
(1) PCR
(2) ELISA
(3) Southern blot
(4) Western blot

141. Placenta has function similar as that of _____ with respect to progesterone secretion.
Choose the option that fills the blank **correctly.**
(1) Liver (2) Corpus luteum
(3) Uterus (4) Fallopian tube

142. The idea that "Certain factors called mutations can bring sudden change even in a span of single generation" was supported by
(1) Darwin (2) Weismann
(3) Ernst Haeckel (4) Hugo de Vries

143. The brain size of Neanderthal man was about
(1) 1400 cc (2) 2400 cc
(3) 1000 cc (4) 500 cc

144. Which of the following is/are non-ionising radiation(s)?
(1) Gamma rays (2) Ultraviolet
(3) X-rays (4) Beta particles

145. Testosterone in human males is produced by
(1) Sertoli cells (2) Leydig cells
(3) Spermatogonia (4) Spermatozoa

146. During cleavage, there is an increase in all of the following, **except**
(1) Number of blastomeres
(2) Size of blastomeres
(3) Nuclear cytoplasmic ratio
(4) DNA content

147. All of the following factors affect enzyme activity directly, **except**
(1) Temperature
(2) Exponential rise in enzyme concentration
(3) Substrate concentration
(4) pH

148. Which of the following is **mismatched** w.r.t. the events that occur in the life cycle of *Plasmodium*?
(1) Release of sporozoites : In salivary glands of humans
(2) Asexual reproduction : In human RBCs and liver cells
(3) Development of gametes : In gut of mosquito
(4) Fertilisation : In gut of mosquito

149. Complete the following analogy:
Scorpion : Book lungs :: Prawn : _____
Choose the **correct** option
(1) Book gills (2) Gills
(3) Coxal gland (4) Green gland

150. *Hirudinaria* differs from *Nereis* in all of the following features, **except**
a. Nephridia to remove excretory waste
b. Development is indirect
c. Dioecious with sexual dimorphism
d. Presence of parapodia
Choose the **correct** option.
(1) a only (2) a, b and c only
(3) a and b only (4) a, b, c and d

151. Closed circulatory system is present in members of
(1) Platyhelminthes (2) Annelida
(3) Aschelminthes (4) Cnidaria

152. The production of uric acid is enhanced by the excessive metabolism of
(1) Carbohydrates (2) Nucleic acid
(3) Lipoprotein (4) Fat

153. Under which of the following conditions appreciable amount of glucose does **not** appear in urine?
a. Diabetes mellitus
b. Diabetes insipidus
c. Glucose concentration exceeds 180 mg/100 ml of blood
d. Excessive secretion of insulin
Choose the **correct** option.
(1) a and b only (2) b and c only
(3) c and d only (4) b and d only

154. Which of the following is an unpaired bone of human body?
(1) Hyoid (2) Maxilla
(3) Palatine bone (4) Nasal bone

155. Foramen ovale is one of the shunts in foetus which connects
(1) Pulmonary artery to pulmonary vein
(2) Right and left atria
(3) Pulmonary artery and aorta
(4) Interventricular septum

156. Choose the **incorrect** statement
(1) The opening of voltage gated Na^+ channels causes depolarisation of axonal membrane
(2) Lubb sound appears at the beginning of ventricular systole
(3) In myocardial infarction, ST segment is depressed and T wave flattens
(4) Angina occur due to conditions that affect the blood flow to heart

157. The joint between prezygapophysis and post zygapophysis of adjacent vertebrae is same as that present between
(1) Atlas and axis (2) Ribs and Sternum
(3) Radius and ulna (4) Tibia and fibula

158. Which of the following is the 6th cranial nerve?
(1) Trochlear nerve (2) Abducens nerve
(3) Occulomotor nerve (4) Optic nerve

159. Choose the **incorrect** option w.r.t. the reaction under the influence of brush border enzymes.
(1) Maltose $\xrightarrow{\text{Maltase}}$ Glucose + Glucose
(2) Dipeptide $\xrightarrow{\text{Dipeptidase}}$ Amino acids
(3) Sucrose $\xrightarrow{\text{Invertase}}$ Glucose + Fructose
(4) Lactose $\xrightarrow{\text{lactase}}$ Maltose + Glucose

160. Auerbach's plexus is present in
(1) Submucosa of intestine
(2) Muscularis layer in anterior part of oesophagus
(3) Muscularis layer of intestine
(4) Mucosal layer of alimentary canal

161. In female cockroach, which of the following structures does **not** help in excretion?
(1) Fat bodies (2) Nephrocyte
(3) Malpighian tubule (4) Uricose gland

162. The maximum amount of air a person can breathe in after forceful expiration is
(1) TV + IRV (2) TLC – RV
(3) ERV + RV (4) TV + ERV

163. Total number of spiracles present in abdomen of cockroach are
(1) 10 pairs (2) 8 pairs
(3) 2 pairs (4) 4 pairs

164. Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds belong to
(1) Hydrolases (2) Lyases
(3) Dehydrogenases (4) Isomerases

165. _____ is made up of a thin mesothelium (epithelium of visceral organs) with some connective tissue. Choose the option which **correctly** fills up the given blank A.
(1) Mucosa (2) Lamina propria
(3) Serosa (4) Submucosa

166. Partial pressure of oxygen in deoxygenated blood is similar to
(1) pCO_2 in tissue
(2) pO_2 in alveoli
(3) pCO_2 in oxygenated blood
(4) pCO_2 in deoxygenated blood

167. All of the following are responsible for increasing blood volume and pressure **except**
(1) Renin
(2) Aldosterone
(3) Angiotensin II
(4) Atrial natriuretic factor

168. Choose the **correct** option w.r.t. the cause of osteoporosis
(1) Accumulation of uric acid crystals
(2) Decreased level of estrogen
(3) Decreased level of PTH
(4) Increased calcium consumption

169. Which of the following is considered as sensory spot of ampulla and responsible for maintenance of dynamic balance of body and posture?
(1) Tympanic membrane
(2) Crista
(3) Helicotrema
(4) Ear ossicles

170. Squamous epithelium is present in the lining of
(1) Alveoli and capillaries
(2) Lining of urinary bladder
(3) Trachea
(4) Stomach

171. Which of the following is a direct method of gene transfer?
(1) Cosmid mediated
(2) Phagemid mediated
(3) Use of disarmed retrovirus
(4) Electroporation

172. Which of the following is an example of dense regular connective tissue?
(1) Skin (2) Epimysium
(3) Ligament (4) Periosteum

173. Chitinous exoskeleton and jointed appendages are features of
(1) Earthworm (2) Cockroach
(3) Sea star (4) *Sycon*

174. Which of the following is **not** a criticism of Natural Selection theory?
(1) Arrival of fittest
(2) Vestigeal organs
(3) Discontinuous variation
(4) Mimicry and protective colouration

175. Choose the **odd** one w.r.t. connecting links
(1) *Peripatus* (2) *Chimaera*
(3) *Neopilina* (4) *Limulus*

176. Read the following characters
(a) Orthognathous face
(b) Cranial capacity of about 1650 cc
(c) Omnivorous
(d) A well developed chin
Characters given above best describe
(1) *Homo erectus*
(2) *Homo sapiens fossilis*
(3) *Homo sapiens neanderthalis*
(4) *Homo habilis*

177. Complete double circulation and amniotic cleidoic eggs are found in
(1) *Balaenoptera* (2) Rabbit
(3) *Neophron* (4) Humans

178. Lymph differs from serum in having
(1) Immunoglobulins (2) WBCs
(3) RBCs (4) Water

179. Cytidine is a
(1) Nitrogenous base
(2) Nucleoside
(3) Nucleotide
(4) Monomer of nucleic acid

180. Which of the following condition does **not** result in shifting of oxyhaemoglobin dissociation curve to right?
(1) High pH
(2) High pCO_2
(3) High H^+ concentration
(4) High temperature

❑ ❑ ❑

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